



USN

NITTE (Deemed to be University)

(Established under Sec.(3) of UGC Act, 1956)

First Semester M.Tech (CBCS) Degree Examinations, December 2025

25VDE111 – ADVANCED DIGITAL SYSTEM DESIGN

(For VDE)

Duration: 3 Hours]

[Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Module – I & Module – II** each and **One full** question from **Module – III**.

Module – I

Marks BT CO* PO*

1. a) Write the state table and draw the state diagram for serial adder. Also write the transition and output tables for a serial binary adder and hence arrive at logical equations.
- b) The direct sum $M_1 + M_2$ of two machines M_1 and M_2 is obtained by combining the tables of the individual machines, as shown in Fig. 1(b), in such a way that each state of the direct sum is denoted by a distinct symbol.
- a) Use the direct sum to determine whether state A of machine M_1 is equivalent to state H of machine M_2 .
- b) Prove that machine M_1 is contained in machine M_2 .
- c) Under what starting conditions are machines M_1 and M_2 equivalent?

10 3 1 3

NS, z			NS, z			NS, z		
PS	x = 0	x = 1	PS	x = 0	x = 1	PS	x = 0	x = 1
A	B, 0	C, 1	E	H, 1	E, 0	A	B, 0	C, 1
B	D, 1	C, 0	F	F, 1	E, 0	B	D, 1	C, 0
C	A, 1	C, 0	G	E, 0	G, 1	C	A, 1	C, 0
D	B, 1	C, 0	H	F, 0	E, 1	D	B, 1	C, 0
						E	H, 1	E, 0
						F	F, 1	E, 0
						G	E, 0	G, 1
						H	F, 0	E, 1

M_1
 M_2
 $M_1 + M_2$

Fig. 1(b): Direct Sum Machine

10 3 2 3

2. a) Machine M_3 shown in Fig. 2(a). Determine its state partition and hence find the reduced Machine M_3 .

NS, z		
PS	x = 0	x = 1
A	E, 0	C, 0
B	C, 0	A, 0
C	B, 0	G, 0
D	G, 0	A, 0
E	F, 1	B, 0
F	E, 0	D, 0
G	D, 0	G, 0

Fig. 2(a): Machine M_3

10 3 2 3

- b) Explain the design of a serial parity-bit generator using synchronous sequential circuits. With the help of a neat state diagram and state table, describe how the circuit ensures even parity for 3-bit input messages. Further, write the logic equations obtained when JK flip-flops are used as memory elements.

10 3 1 3

3. a) Analyze the synchronous circuit given in Fig. 3(a)
- Write down the excitation and output functions.
 - Form the excitation and state tables.
 - Give a word description of the circuit operation

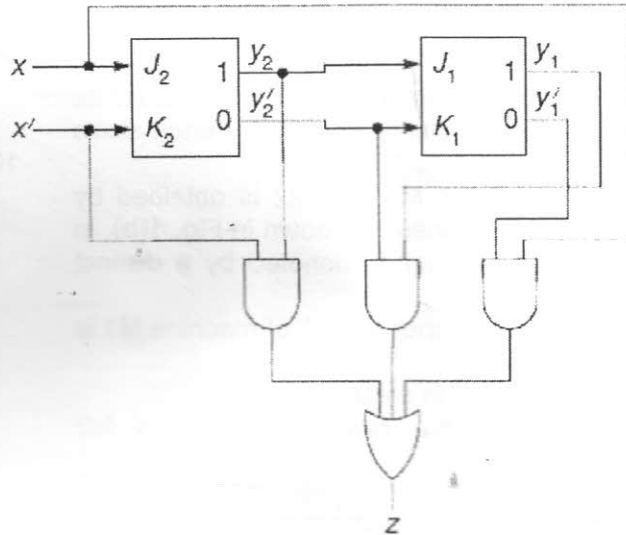


Fig. 3(a): Synchronous circuit

10 3 1 3

- b) Draw the merger graph and compatibility graph for Machine M4 shown in Fig. 3(b). Hence find the minimal machine.

PS	NS, z			
	I_1	I_2	I_3	I_4
A	—	—	E, 1	—
B	C, 0	A, 1	B, 0	—
C	C, 0	D, 1	—	A, 0
D	—	E, 1	B, —	—
E	B, 0	—	C, —	B, 0

Fig. 3(b): Machine M4

10 3 2 3

Module – II

4. a) Consider the primitive flow table shown in Table 4a. Explain the method to reduce the flow table and hence find the reduced flow table. Write the equivalent flow table with state assignment. Also find the implication table corresponding to this flow table.

10 3 3 3

	x_1x_2			
	00	01	11	10
S_0	$\textcircled{S_0}, 0$	—	—	S_1
S_1	S_2	—	—	$\textcircled{S_1}, 1$
S_2	$\textcircled{S_2}, 1$	S_5	—	S_3
S_3	S_4	—	—	$\textcircled{S_3}, 1$
S_4	$\textcircled{S_4}, 1$	S_6	—	—
S_5	S_0	$\textcircled{S_5}, 1$	—	—
S_6	S_2	$\textcircled{S_6}, 1$	—	—

Table 4a: Primitive Flow Table

- b) Construct the flow table of a sequential circuit that has two inputs x_1x_2 and one output Z ; the circuit produces an output of 1 only if the input sequence $x_1x_2 = 10, 11, 01$ is received in that order. All other sequences produce an output of 0.

10 3 3 3

- a) Discuss the concept of a critical race with reference to the flow table 1 given in Table 5a. Assume that the initial state is A with both inputs x_1 and x_2 set to 0. Analyze the behavior of the circuit when the input x_2 changes to 1, and explain how this illustrates the concept of a critical race.

	x_1x_2			
	00	01	11	10
A	$\textcircled{A}$	C	$\textcircled{A}$	C
B	A	$\textcircled{B}$	$\textcircled{B}$	D
C	D	$\textcircled{C}$	B	$\textcircled{C}$
D	$\textcircled{D}$	B	A	$\textcircled{D}$

Table 5a: Flow table 1

10 3 3 3

- b) Draw the Karnaugh map and the corresponding gate-level circuit for the function $T(x, y, z) = \Sigma(2, 3, 5, 7)$. Explain the concept of a static hazard that occurs due to a single input change. Modify the circuit to obtain a hazard-free implementation.

10 3 3 3

6. a) Consider the flow table given in Table 6a. Arrive at a reduced flow table using a single transition time assignment method.

State	x_1x_2			
	00	01	11	10
S_0	(S_0)	S_1	(S_0)	(S_0)
S_1	S_2	(S_1)	(S_1)	S_0
S_2	(S_2)	S_3	S_1	(S_2)
S_3	S_4	(S_3)	(S_3)	S_2
S_4	(S_4)	S_1	S_3	(S_4)

Table 6a: Flow table 2

10 3 3 3

- b) What are static and dynamic hazards? With the help of a circuit shown in Fig. 6(b), explain hazards in combinational design.

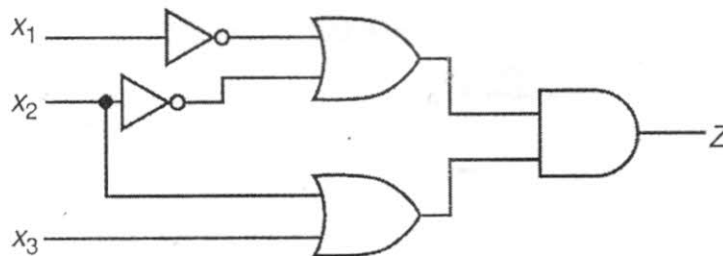


Fig. 6(b): Combinational circuit

10 3 3 3

Module – III

7. a) Explain the terms bound line, free line, head line and technique multiple backtracks used in FAN algorithm. Illustrate the application of the FAN algorithm by deriving a test for the fault Z s-a-0 in Fig. 7(a) considering any sequential circuit as an example.

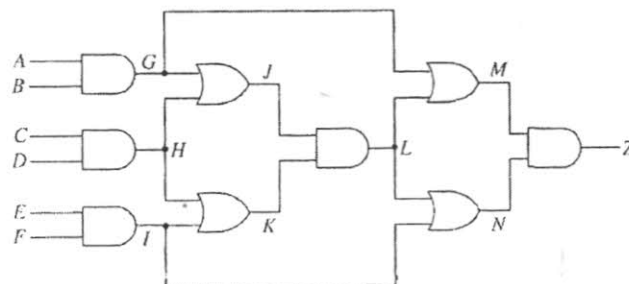


Fig. 7(a): Circuit under test

10 3 5 3

b) Write the different steps involved in the PODEM algorithm. Illustrate the application of PODEM by deriving a test for fault I s-a-1 in the circuit shown in Fig. 7(b).

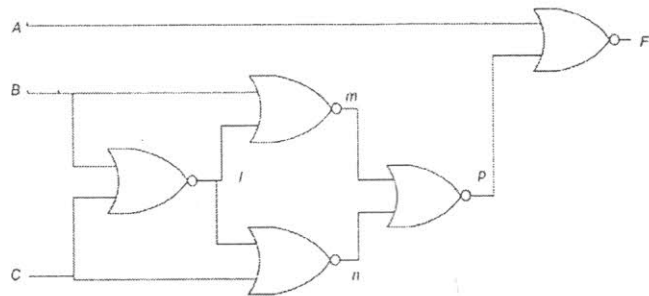


Fig. 7(b): Circuit under test

8. a) With the help of an example for each, Explain 3 types of cubes used in the D algorithm.
- b) Illustrate the test generation process by deriving a test sequence for the fault α s-a-0 in the sequential circuit shown in Fig. 8(b).

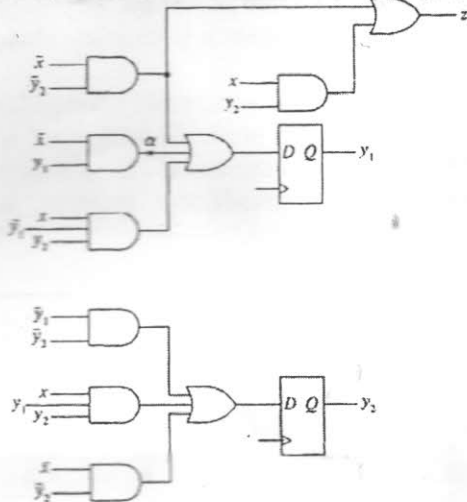


Figure 8b: A sequential circuit

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
